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Vehicle rear structure and trailer coupling structure of vehicle

Abstract

A vehicle rear structure includes an interposed member and an attachment part. The interposed member is disposed at a center area in a vehicle width direction of the vehicle under a floor panel of a vehicle body of the vehicle, and interposed between at least one under-floor component of the vehicle and the vehicle body. The attachment part is provided for a hitch member for a trailer. The attachment part is disposed at a rear end of the interposed member and at a center area of the interposed member in the vehicle width direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority from Japanese Patent Application No. 2021-151856 filed on Sep. 17, 2021, the entire contents of which are hereby incorporated by reference.

BACKGROUND

(2) The disclosure relates to a vehicle rear structure having an attachment part for a hitch member for a trailer, and a trailer coupling structure of a vehicle provided with the vehicle rear structure.

(3) There is a known trailer coupling structure of a vehicle in which a fixing bar as a hitch member extending in a vehicle width direction is fixed to a coupling bar permanently affixed to a vehicle body frame at a rear end of the vehicle and extending in the vehicle width direction, and a front end portion of a base of a trailer is pivotally coupled to a hook disposed on the fixing bar (for example, see Japanese Unexamined Patent Application Publication (JP-A) No. 2000-203467). There is also a known trailer coupling structure in which a hitch member extending in a vehicle width direction is fixed to a side frame extending in a front-rear direction at a rear end of a vehicle.

SUMMARY

(4) An aspect of the disclosure provides a rear structure of a vehicle. The rear structure includes an interposed member and an attachment part. The interposed member is disposed at a center area in a vehicle width direction of the vehicle under a floor panel of a vehicle body of the vehicle, and interposed between at least one under-floor component of the vehicle and the vehicle body. The attachment part is provided for a hitch member for a trailer. The attachment part is disposed at a rear end of the interposed member at a center area of the interposed member in the vehicle width direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate example embodiments and, together with the specification, serve to describe the principles of the disclosure.

(2) FIG. 1 is a schematic diagram illustrating a vehicle and a trailer according to an embodiment of the disclosure.

(3) FIG. 2 is a bottom view illustrating a vehicle rear structure when a hitch member is attached.

(4) FIG. 3 is a back cross-sectional view illustrating the vehicle rear structure when the hitch member is attached.

(5) FIG. 4 is a cross-sectional view taken along a line A-A of FIG. 3.

(6) FIG. 5 is a back cross-sectional view illustrating the vehicle rear structure before the hitch member is attached.

DETAILED DESCRIPTION

(7) In the trailer coupling structure of the vehicle described in JP-A No. 2000-203467, it is necessary to secure an attachment space for a hitch member on a rear end side of a vehicle body. Thus, it is difficult to attach a hitch member to a vehicle having a small amount of rearward overhang from a rear axle. There is also a problem in that when the hitch member is attached to the rear end side of the vehicle body, deformation, cracking, and the like are likely to occur in the vehicle body.

(8) It is desirable to provide a vehicle rear structure in which a hitch member can be simply and easily attached even in a vehicle having a small amount of overhang from a rear axle, and deformation, cracking, and the like are less likely to occur in a vehicle body, and a trailer coupling structure of a vehicle provided with the vehicle rear structure.

(9) In the following, an embodiment of the disclosure is described in detail with reference to the accompanying drawings. Note that the following description is directed to an illustrative example of the disclosure and not to be construed as limiting to the disclosure. Factors including, without limitation, numerical values, shapes, materials, components, positions of the components, and how the components are coupled to each other are illustrative only and not to be construed as limiting to the disclosure. Further, elements in the following example embodiment which are not recited in a most-generic independent claim of the disclosure are optional and may be provided on an as-needed basis. The drawings are schematic and are not intended to be drawn to scale. Throughout

the present specification and the drawings, elements having substantially the same function and configuration are denoted with the same numerals to avoid any redundant description.

(10) FIG. 1 is a schematic diagram illustrating a vehicle and a trailer according to the embodiment of the disclosure; FIG. 2 is a bottom view illustrating a vehicle rear structure when a hitch member is attached; FIG. 3 is a back cross-sectional view illustrating the vehicle rear structure when the hitch member is attached; FIG. 4 is a cross-sectional view taken along a line A-A of FIG. 3; and FIG. 5 is a back cross-sectional view illustrating the vehicle rear structure before the hitch member is attached.

(11) As illustrated in FIG. 1, a vehicle **100** tows a trailer **200** coupled to a rear portion of the vehicle **100**. The vehicle **100** and the trailer **200** are coupled by a hitch ball **110** and a coupler **210**. The hitch ball **110** is disposed at an upper portion of a ball mount **115**, on a rear end side, extending in a front-rear direction. The coupler **210** is disposed at a front end of the trailer **200** to protrude forward, and receives the hitch ball **110** from an upper side at a front end. A coupling portion between the hitch ball **110** and the coupler **210** is the same as a coupling portion in the related art, and thus will not be described in detail here.

(12) The vehicle **100** has two axles, and a pair of left and right front wheels **120** and a pair of left and right rear wheels **125**, that is, a total of four wheels are grounded. On the other hand, the trailer **200** has one axle, and a pair of left and right wheels **220** are grounded. In the present embodiment, a power unit **130** is disposed on a front side of the vehicle **100**, and power is transmitted to the front wheels **120** while power is also transmitted to the rear wheels **125** via a drive shaft **135**. A rear differential **140** that allows a difference in rotation between the rear wheels **125** is disposed at a rear end of the drive shaft **135**. As illustrated in FIG. 2, an axle **145** that transmits the power to the rear wheels **125** is coupled to the rear differential **140**.

(13) As illustrated in FIG. 2, the rear differential **140** is supported by a suspension frame **150**. Lower arms **160** and upper arms **161** of a rear suspension are attached to the suspension frame **150**. In the present embodiment, the suspension frame **150** forms an interposed member disposed at a center area in a vehicle width direction under a floor panel **171** of a vehicle body **170** and interposed between at least one under-floor component and the vehicle body **170**. In the present embodiment, the under-floor components are a housing of the rear differential **140**, and the lower arms **160** and the upper arms **161** of the rear suspension.

(14) In the present embodiment, the suspension frame **150** includes a first cross member **151** extending in a left-right direction, a second cross member **152** extending in the left-right direction behind the first cross member **151**, and a pair of left and right side members **153** having front and rear ends coupled to the first cross member **151** and the second cross member **152** and extending in the front-rear direction. In the present embodiment, a rear end of the housing of the rear differential **140** is supported by the second cross member **152**.

(15) As illustrated in FIG. 3, the first cross member **151** and the second cross member **152** are fixed to the vehicle body **170** of the vehicle **100** via rubber mounts **180**. The vehicle body **170** includes the floor panel **171** and a pair of left and right side frames **172** disposed on a lower surface of the floor panel **171**. The floor panel **171** and the side frames **172** form a closed cross section. In the present embodiment, the first cross member **151** and the second cross member **152** are fixed to the side frames **172** via the rubber mounts **180**.

(16) The lower arms **160** of the rear suspension are rotatably attached to knuckles (not illustrated) of the rear wheels and the second cross member **152**. The upper arms **161** of the rear suspension are rotatably attached to the knuckles of the rear wheels and the side members **153**. The rear suspension includes struts **162** that are each interposed between the floor panel **171** and the lower arm **160** and each include a spring and a shock absorber.

(17) As illustrated in FIG. 2, a hitch member **190** can be attached to a back surface of the second cross member **152**. The hitch member **190** includes a member main body **191** that horizontally extends in the front-rear direction, and a bracket part **192** that fixes the member main body **191** to

the second cross member **152**. The member main body **191** is formed in a rectangular tube shape, and receives a front end of the ball mount **115** from an opening **191a** at a rear end. In the present embodiment, as illustrated in FIG. **4**, a pin hole **191b** is formed in each side surface of the member main body **191**. The ball mount **115** and the member main body **191** can be fixed by inserting a pin (not illustrated) into the ball mount **115** and the member main body **191**, when the ball mount **115** is inserted into the member main body **191**.

(18) As illustrated in FIG. **2**, the bracket part **192** is formed in a substantially hat-shaped plate shape in a bottom view, and includes a holding portion **192a** that holds the member main body **191** at a predetermined position in the front-rear direction, a pair of left and right extending portions **192b** that extend forward from left and right ends of the holding portion **192a** while being inclined to left and right outer sides, and fixing portions **192c** that respectively extend from front ends of the extending portions **192b** to the left and right outer sides along the back surface of the second cross member **152**. The fixing portions **192c** of the bracket part **192** are fixed to the second cross member **152** by bolts **193**.

(19) In the present embodiment, as illustrated in FIG. **5**, a hitch member attachment part of the second cross member **152** includes a seat surface **152a** of the fixing portions **192c** of the bracket part **192** that is formed on the back surface of the second cross member **152**, and female screw portions **152b** that are formed on the seat surface **152a** and screwed with the bolts **193**. That is, the seat surface **152a** and the female screw portions **152b** form an attachment part for the hitch member **190** for the trailer **200** that is disposed at a rear end of the suspension frame **150** at the center area in the vehicle width direction.

(20) According to the vehicle rear structure configured as described above, the hitch member **190** for the trailer **200** is attached to the attachment part at the rear end of the suspension frame **150** disposed under the floor panel **171**. The member main body **191** of the hitch member **190** extends in the front-rear direction under the floor panel **171** when the hitch member **190** is attached to the suspension frame **150**, and the coupler **210** of the trailer **200** is coupled to the hitch ball **110** of the ball mount **115** coupled to a rear end side of the hitch member **190**. Accordingly, the hitch member **190** can be attached to the vehicle body at a position lower than that in the related art in which a hitch member is attached to a vehicle body. Thus, the center of gravity of the vehicle **100** can be lowered when the trailer **200** is coupled, so that the motion performance of the vehicle **100** can be improved, and rollover of the trailer **200** can be prevented.

(21) Further, when a load is input from a trailer **200** side to a vehicle **100** side, the load is transmitted from the hitch member **190** to the vehicle body **170** via the suspension frame **150**, and thus it is possible to reduce vibration and noise in a vehicle cabin as compared with the vehicle in the related art in which a load is directly transmitted from the hitch member to the vehicle body. In the present embodiment, since the rubber mounts **180** are interposed between the suspension frame **150** and the vehicle body **170**, the vibration and the noise caused by the trailer **200** are hardly transmitted to a vehicle cabin side. Further, it is not necessary to secure an attachment space for the hitch member **190** on a rear end side of the vehicle body **170**, so that the degree of freedom in designing the rear end side of the vehicle body **170** increases. Also, the hitch member **190** can be simply and easily attached to the vehicle **100** having a small amount of rearward overhang from the axle **145** on the rear side.

(22) The suspension frame **150** under the floor panel **171** is exposed outside the vehicle. Therefore, when attaching the hitch member **190** to the vehicle, unlike the related art in which the hitch member is attached to a rear end of a frame of the vehicle body, it is not necessary to remove a bumper, and the workability of attaching the hitch member **190** can be improved. In addition, since the hitch member **190** is attached to the vehicle body **170** via the suspension frame **150** having a relatively high rigidity, deformation, cracking, and the like are less likely to occur in the vehicle body as compared with the related art in which the hitch member is attached to the rear end of the frame of the vehicle body. In addition, it is possible to reduce the weight and cost due to the

simplified component configuration. Since the suspension frame **150** has the relatively high rigidity, the suspension frame **150** is suitable for attaching the hitch member.

(23) In the above-described embodiment, the interposed member is interposed between the vehicle body **170** and components of both the rear suspension and the rear differential **140**. However, the interposed member may be interposed between the vehicle body **170** and the components of one of the rear suspension and the rear differential **140**. For example, the interposed member may be a bracket that supports the components of the rear differential **140** but does not support the components of the suspension. Since the bracket of the rear differential **140** has a relatively high rigidity, the bracket is suitable for attaching the hitch member. The under-floor component may be a component other than the components of the rear suspension and the rear differential, and may be, for example, a component of a motor unit that drives the rear wheels. Since a bracket of the motor unit has a relatively high rigidity, the bracket is suitable for attaching the hitch member.

(24) Although the embodiment of the disclosure has been described above, the embodiment described above does not limit the disclosure according to the claims. It is also noted that not all the combinations of the features described in the embodiment are necessary for solving the problem of the disclosure.

(25) According to the disclosure, the hitch member can be simply and easily attached even in the vehicle having a small amount of overhang from the axle on the rear side, and deformation, cracking, and the like are less likely to occur in the vehicle body.

Claims

1. A rear structure of a vehicle, the rear structure comprising: an interposed member disposed at a center area in a vehicle width direction of the vehicle under a floor panel of a vehicle body of the vehicle, and interposed between at least one under-floor component of the vehicle and the vehicle body; and an attachment part for a hitch member for a trailer, the attachment part being disposed at a rear end of the interposed member and at a center area of the interposed member in the vehicle width direction, wherein the attachment part extends parallel to a bottom surface of the interposed member and a top surface of the interposed member.
2. The vehicle rear structure according to claim 1, wherein the at least one under-floor component comprises a component of a rear suspension.
3. The vehicle rear structure according to claim 1, wherein the at least one under-floor component comprises a component of a rear differential.
4. The vehicle rear structure according to claim 2, wherein the at least one under-floor component comprises multiple under-floor components, and wherein the under-floor components comprise a component of a rear differential.
5. The vehicle rear structure according to claim 1, wherein the at least one under-floor component comprises a component of a motor unit.
6. The vehicle rear structure according to claim 2, wherein the at least one under-floor component comprises multiple under-floor components, and wherein the under-floor components comprise a component of a motor unit.
7. The vehicle rear structure according to claim 3, wherein the at least one under-floor component comprises multiple under-floor components, and wherein the under-floor components comprise a component of a motor unit.
8. The vehicle rear structure according to claim 4, wherein the under-floor components comprise a component of a motor unit.
9. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 1; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch

member is coupled to a ball mount comprising a hitch ball.

10. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 2; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch member is coupled to a ball mount comprising a hitch ball.

11. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 3; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch member is coupled to a ball mount comprising a hitch ball.

12. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 4; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch member is coupled to a ball mount comprising a hitch ball.

13. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 5; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch member is coupled to a ball mount comprising a hitch ball.

14. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 6; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch member is coupled to a ball mount comprising a hitch ball.

15. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 7; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch member is coupled to a ball mount comprising a hitch ball.

16. A trailer coupling structure of a vehicle, the trailer coupling structure comprising: the rear structure according to claim 8; and a hitch member comprising a member main body horizontally extending in a front-rear direction of the vehicle, wherein a front end of the hitch member is attached to the attachment part of the interposed member, and wherein a rear end of the hitch member is coupled to a ball mount comprising a hitch ball.

17. The vehicle rear structure according to claim 1, wherein the attachment part is attached to the interposed member by bolts.

18. The vehicle rear structure according to claim 2, further comprising a rubber mount that fixes the interposed member to the vehicle body.

19. A rear structure of a vehicle, the rear structure comprising: an interposed member disposed at a center area in a vehicle width direction of the vehicle under a floor panel of a vehicle body of the vehicle, and interposed between at least one under-floor component of the vehicle and the vehicle body; and an attachment part for a hitch member for a trailer, the attachment part being disposed at a rear end of the interposed member and at a center area of the interposed member in the vehicle width direction, wherein the at least one under-floor component comprises a component of a rear differential.

20. A rear structure of a vehicle, the rear structure comprising: an interposed member disposed at a center area in a vehicle width direction of the vehicle under a floor panel of a vehicle body of the vehicle, and interposed between at least one under-floor component of the vehicle and the vehicle

body; and an attachment part for a hitch member for a trailer, the attachment part being disposed at a rear end of the interposed member and at a center area of the interposed member in the vehicle width direction, wherein the at least one under-floor component comprises a component of a rear suspension, wherein the at least one under-floor component comprises multiple under-floor components, and wherein the under-floor components comprise a component of a rear differential.
